

Meta-analyses of lignans and enterolignans in relation to breast cancer risk.



BACKGROUND: Epidemiologic studies that examined whether lignans, the most important class of phytoestrogens in the Western diet, protect against breast cancer have yielded inconsistent results. OBJECTIVE: In this study, we conducted meta-analyses on the association between lignans and breast cancer risk. DESIGN: We performed a systematic MEDLINE search to identify epidemiologic studies published between 1997 and August 2009. We calculated pooled risk estimates (REs) for total lignan exposure, dietary lignan intake, enterolignan exposure, and blood or urine concentrations of enterolactone and according to menopausal and estrogen receptor (ER) status of tumors. RESULTS: We included 21 studies (11 prospective cohort studies and 10 case-control studies) in the meta-analyses. Lignan exposure was not associated with an overall breast cancer risk (RE: 0.92; 95% CI: 0.81, 1.02; P for heterogeneity = 0.004). However, in postmenopausal women, high lignan intake was associated with a significant reduced risk of breast cancer (13 studies; RE: 0.86; 95% CI: 0.78, 0.94; P for heterogeneity = 0.32). Breast cancer risk was also inversely associated with enterolignan exposure (4 studies; RE: 0.84; 95% CI: 0.71, 0.97) but not with blood or urine enterolactone concentrations. The associations were not significantly different between ER-status subgroups (6 studies). CONCLUSIONS: High lignan exposure may be associated with a reduced breast cancer risk in postmenopausal women. Additional work is warranted to clarify the association between lignan exposure and breast cancer risk.